Math 2C03 - Assignment 1

Fall 2026

This is one generated instance of the assignment. Your Childsmath questions may have different numerical values or choice orders. Submit your answers in Childsmath.

Question 1

Classify the following differential equation according to type, order, and linearity, in that order.

$$-5x\left(\frac{dy}{dx}\right)^2 + \sin(20x)y = e^{4x}.$$

(A) ordinary, first order, non-linear	(B) ordinary, second order, non-linear
(C) ordinary, first order, linear	(D) ordinary, second order, linear
(E) partial, first order, linear	(F) partial, second order, non-linear
(G) partial, second order, linear	(H) partial, first order, non-linear

Question 2

Find the two values of m so that the function $y = x^m$ is a solution to the following differential equation.

$$x^2y'' - 16xy' + 72y = 0.$$

Separate your answers with a comma. If there is only one value of m then write it twice, e.g., $12, 12$.

Question 3

For which of the following points does the existence and uniqueness theorem stated in lecture 2 guarantee that the differential equation

$$y' = e^{x^2}\sqrt{36 - y^2}$$

possesses a unique solution through the given point?

(i) $(4, 6)$

(ii) $(-3, 3)$

(iii) $(5, 5)$

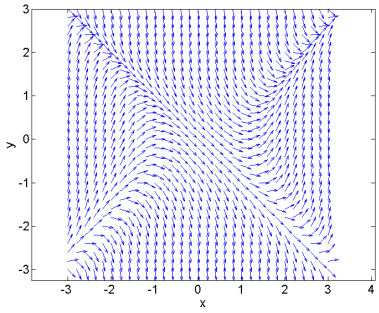
(A) (i) only	(B) (ii) only
(C) (iii) only	(D) none of them
(E) all of them	(F) (i) and (iii) only
(G) (ii) and (iii) only	(H) (i) and (ii) only

Question 4

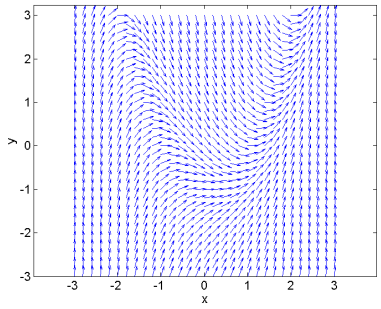
Which of the following is a direction field for the differential equation

$$\frac{dy}{dx} = x^2y^2 - 1 \ ?$$

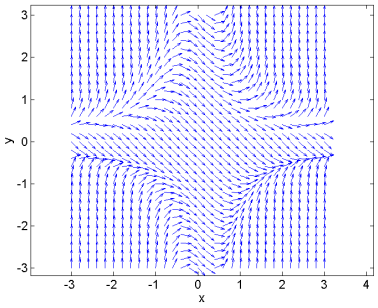
(A)



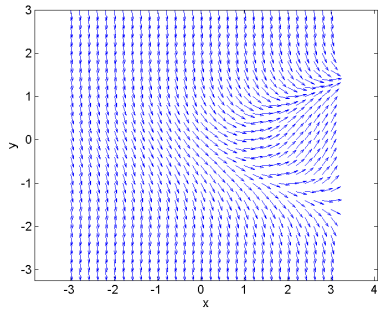
(B)



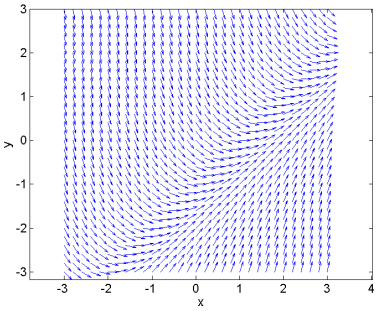
(C)



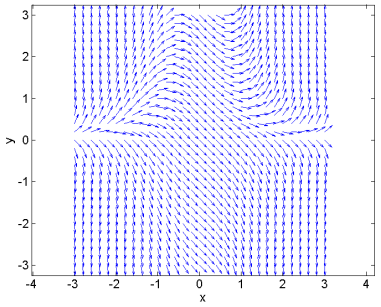
(D)



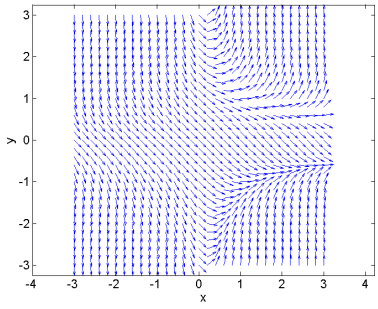
(E)



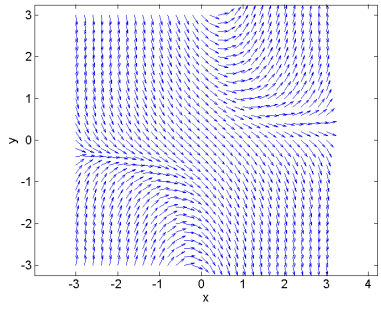
(F)



(G)



(H)

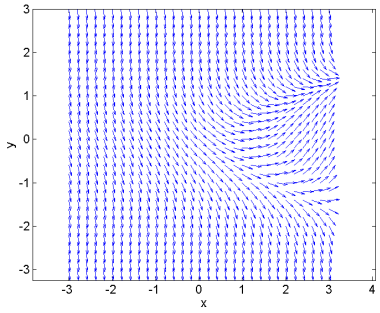


Question 5

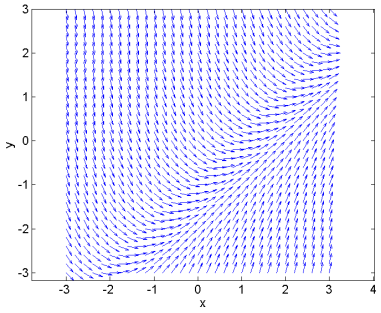
Which of the following is a direction field for the differential equation

$$\frac{dy}{dx} = x - y - 1 \ ?$$

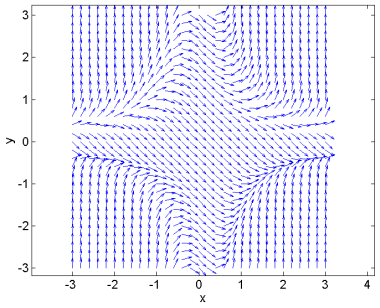
(A)



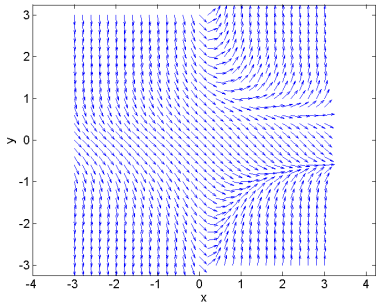
(B)



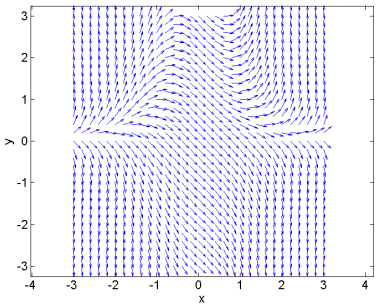
(C)



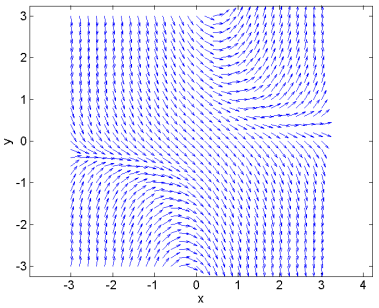
(D)



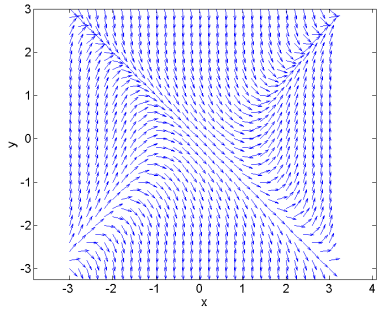
(E)



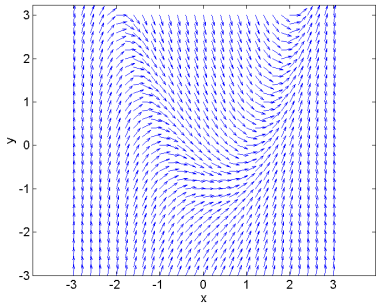
(F)



(G)



(H)



Question 6

Let $y(x)$ be the solution to the following initial value problem.

$$y^2\,dx - \sec^2(9x)\,dy = 0, \qquad y(0) = 8.$$

Find $y(\pi)$.

Question 7

A model for a certain population $P(t)$ is given by the initial value problem

$$\frac{dP}{dt} = P(10^{-3} - 10^{-10}P), \qquad P(0) = 200000,$$

where t is measured in months.

(a) What is the limiting value of the population?

(b) At what time (i.e., after how many months) will the population be equal to one sixth of the limiting value in (a)? Do not round any numbers for this part. Your work should be all symbolic.